

KRISH KUMAR

Full-Stack Software Engineer — React, Node.js & REST APIs

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Summary

Engineering graduate (2026) who builds full-stack web applications end to end across the MERN stack — React single-page front ends consuming Node.js and Express REST APIs backed by MongoDB. Has implemented JWT authentication with access and refresh tokens, bcrypt password hashing, cookie-based sessions, file upload handling, and indexed document data models with aggregation queries. Also builds in Java with Spring Boot and Python with FastAPI, which gives a comparative view of REST API design across ecosystems. Writes structured, layered code rather than single-file servers, debugs across the front-end and back-end boundary, and learns new tooling fast and unaided. 500+ DSA problems solved.

Technical Skills

Languages: JavaScript (ES6+), Java, Python, SQL, HTML5, CSS3

Frontend: React (v19, hooks, component architecture), React Router, Vite, Axios, responsive UI

Backend: Node.js, Express.js (v5), REST API design, middleware, routing, CORS, cookie-parser, dotenv, Spring Boot, FastAPI

Databases: MongoDB with Mongoose — schema design, indexed fields, document references, aggregation pipelines; MySQL, SQL, Hibernate/JPA

Authentication & Security: JWT access & refresh tokens, bcrypt password hashing, cookie-based sessions, role-based access control, environment-based secrets

Version Control & Tooling: Git, GitHub, pull requests, npm, Postman, VS Code, IntelliJ IDEA, Docker, Docker Compose, GitHub Actions

File Upload & Media: multer multipart handling, ImageKit, Cloudinary cloud storage and delivery

Concepts & Foundations: System design (scalability, load balancing, caching, sharding & replication, microservices, rate limiting), Data Structures & Algorithms, OOP, DBMS

Projects

Reel-Style Video Platform — Full-Stack MERN Application — React 19, Express 5, MongoDB

- Built a full-stack application end to end — a React 19 single-page front end (Vite, React Router, Axios) consuming a Node.js and Express 5 REST API backed by MongoDB via Mongoose.
- Implemented JWT authentication with bcrypt password hashing and cookie-based token handling, and configured CORS so a separately hosted front end can call the API securely.
- Handled video upload and delivery with multer for multipart processing and ImageKit for cloud storage and transformation, keeping media off the application server; structured the repository as independent frontend and backend services with environment-variable configuration. Source: github.com/krishkumar6/zomato-reelstyle

Video Platform Backend — REST API & Data Modelling — Node.js, Express 5, MongoDB, JWT

- Built a Node.js and Express REST API organised into controllers, routes, models, middlewares, utils, and db layers using ES modules — a maintainable structure rather than a single-file server.
- Modelled Users, Videos, and Subscriptions in MongoDB with Mongoose, using indexed fields for lookup performance and document references for relationships such as a user's watch history, with paginated aggregation queries for efficient list endpoints.
- Implemented JWT access and refresh token authentication with a bcrypt pre-save hook hashing passwords before persistence, plus media upload handling via multer with Cloudinary storage. Source: github.com/krishkumar6/youtube-backend

Healthcare Appointment Platform — Full-Stack with Access Control — Java, Spring Boot, React, MySQL

- Built a full-stack platform with a Spring Boot REST API, a React front end, and MySQL, integrating a separate Python service over HTTP and surfacing its output in a prioritised dashboard.
- Implemented JWT authentication and role-based access control across three user roles so authorisation is enforced at the API layer rather than in the UI, and debugged defects across the service boundary with payload validation and downstream failure handling.

Education

G.L. Bajaj Institute of Technology and Management

Nov 2022 – May 2026

B.Tech, Electronics & Communication Engineering — 70%

Coursework: Data Structures & Algorithms, Object-Oriented Programming, DBMS, Operating Systems, Computer Networks.

Achievements

- Solved 500+ problems on LeetCode — arrays, trees, graphs, and dynamic programming; strong debugging and edge-case reasoning.
- Write a public engineering blog at krishkumar6.github.io explaining system design and software engineering concepts. Self-taught the full stack across two ecosystems, shipping five projects solo.